

(LSTM)

Vocabulary

TASK 1

the = One-hot encoded X. SkipGram
vector of 'the' Target weights
 (1×8) Matrix (8×4)
 (1×4)

the =

3	2	3	4
---	---	---	---

 laptops =

2	2	0	0
---	---	---	---

students =

0	3	1	4
---	---	---	---

 books =

4	2	4	2
---	---	---	---

opened =

3	2	0	2
---	---	---	---

 exams =

2	2	4	3
---	---	---	---

their =

1	3	0	2
---	---	---	---

 minds =

4	3	4	2
---	---	---	---

2. Embeddings:- Context Words

the	=	1	0	1	4
students	=	1	1	1	1
opened	=	2	0	2	2
their	=	0	1	2	1
laptops	=	1	4	0	1
books	=	4	0	4	4
exams	=	0	1	3	3
minds	=	1	2	3	0

the = One hot encoded X Skig Gram Context
 vector of 'the' Weights Matrix
 (1×8) (4×8)
 (1×4)

3. Embeddings:- Average

the	=	$\frac{3+1}{2}$	$\frac{2+0}{2}$	$\frac{3+1}{2}$	$\frac{4+4}{2}$
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the	=	2	1	2	4
students	=	0.5	2	1	2.5
opened	=	2.5	1	1	2
their	=	0.5	2	1	1.5
laptops	=	1.5	3	0	0.5
books	=	4	1	4	3
exams	=	1	1.5	3.5	3
minds	=	2.5	2.5	3.5	1

TASK 2 USING TARGET WORD EMBEDDINGS

the	=	3	2	3	4	→ n_1
students	=	0	3	1	4	→ n_2
opened	=	3	2	0	2	→ n_3
their	=	1	3	0	2	→ n_4

At time = 1

$$h_{t-1} = h_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad c_{t-1} = c_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\underline{I} = [h_{t-1}, n_t]$$

$$\underline{I} = [h_0, n_1]$$

$$\underline{I} = [0 \ 0 \ 0 \ 0 \ 3 \ 2 \ 3 \ 4]$$

$$f_1 = \text{sigmoid}(W_f * \underline{I} + B_f)$$

$$f_1 = \text{sigmoid} \left[\begin{bmatrix} 2 & 2 & 2 & 6 & 0 & 6 & 4 & 6 \\ 4 & 1 & 1 & 0 & 2 & 0 & 3 & 5 \\ 1 & 2 & 1 & 3 & 4 & 5 & 1 & 6 \\ 2 & 2 & 2 & 4 & 1 & 1 & 5 & 2 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 3 \\ 2 \\ 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right]$$

$$(4 \times 8) \times (8 \times 1) = (4 \times 1)$$

$$f_1 = \text{sigmoid} \left[\begin{bmatrix} 48 \\ 35 \\ 49 \\ 28 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right] = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_1 = \text{sigmoid}(W_i * I + B_i)$$

$$i_1 = \text{sigmoid} \left[\begin{bmatrix} 6 & 5 & 5 & 0 & 1 & 5 & 0 & 1 \\ 3 & 2 & 2 & 6 & 2 & 1 & 5 & 6 \\ 3 & 2 & 1 & 3 & 0 & 2 & 3 & 2 \\ 4 & 4 & 1 & 5 & 4 & 1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right]$$

$$i_1 = \text{sigmoid} \begin{bmatrix} 17 \\ 47 \\ 22 \\ 32 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$O_1 = \text{sigmoid}(W_o * I + B_o)$$

$$O_1 = \text{sigmoid} \left[\begin{bmatrix} 5 & 4 & 3 & 5 & 1 & 3 & 2 & 4 \\ 4 & 2 & 2 & 1 & 5 & 0 & 1 & 2 \\ 3 & 0 & 4 & 0 & 2 & 1 & 1 & 5 \\ 4 & 1 & 5 & 3 & 5 & 3 & 3 & 6 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right]$$

$$O_1 = \text{sigmoid} \begin{bmatrix} 37 \\ 30 \\ 31 \\ 68 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$c_1^{\sim} = \tanh(W_c * I + b_c)$$

$$c_1^{\sim} = \tanh \left[\begin{bmatrix} 0 & 5 & 6 & 0 & 5 & 4 & 0 & 3 \\ 0 & 1 & 5 & 6 & 0 & 3 & 3 & 6 \\ 0 & 0 & 5 & 5 & 0 & 5 & 3 & 4 \\ 2 & 0 & 6 & 5 & 5 & 5 & 6 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right]$$

$$c_1^{\sim} = \tanh \begin{bmatrix} 35 \\ 41 \\ 35 \\ 53 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$C_1 = f_1 * C_0 + i_1 * C_1^{\sim}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_1 = o_1 * C_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

At time = 2

$$I = [h_1, n_2]$$

$$I = [1 \ 1 \ 1 \ 1 \ 0 \ 3 \ 1 \ 4]$$

$$f_2 = \text{sigmoid} \left[\begin{array}{c|c|c} \begin{bmatrix} 2 & 2 & 2 & 6 & 0 & 6 & 4 & 6 \\ 4 & 1 & 1 & 0 & 2 & 0 & 3 & 5 \\ 1 & 2 & 1 & 3 & 4 & 5 & 1 & 6 \\ 2 & 2 & 2 & 4 & 1 & 1 & 5 & 2 \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} & \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \end{array} \right]$$

$$f_2 = \begin{bmatrix} 58 \\ 31 \\ 48 \\ 26 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_2 = 6 \left[\begin{array}{c|c|c} \begin{bmatrix} 6 & 5 & 5 & 0 & 1 & 5 & 0 & 1 \\ 3 & 2 & 2 & 6 & 2 & 1 & 5 & 6 \\ 3 & 2 & 1 & 3 & 0 & 2 & 3 & 2 \\ 4 & 4 & 1 & 5 & 4 & 1 & 4 & 0 \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \end{array} \right]$$

$$i_2 = \begin{bmatrix} 35 \\ 45 \\ 27 \\ 27 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$O_2 = \begin{bmatrix} 5 & 4 & 3 & 5 & 1 & 3 & 2 & 4 \\ 4 & 2 & 2 & 1 & 5 & 0 & 1 & 2 \\ 3 & 0 & 4 & 0 & 2 & 1 & 1 & 5 \\ 4 & 1 & 5 & 3 & 5 & 3 & 3 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 3 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix}$$

$$O_2 = \begin{bmatrix} 54 \\ 39 \\ 38 \\ 75 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{C}_2 = \tanh \begin{bmatrix} 0 & 5 & 6 & 0 & 5 & 4 & 0 & 3 \\ 0 & 1 & 5 & 6 & 0 & 3 & 3 & 6 \\ 0 & 0 & 5 & 5 & 0 & 5 & 3 & 4 \\ 2 & 0 & 6 & 5 & 5 & 5 & 6 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 3 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix}$$

$$\tilde{C}_2 = \tanh \begin{bmatrix} 46 \\ 53 \\ 45 \\ 66 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_2 = f_2 * C_1 + i_2 * \tilde{C}_2$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

$$h_2 = O_2 + C_2$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

At time = 3

$$I_3 = [h_2, n_3]$$

$$I_3 = [2 \ 2 \ 2 \ 2 \ 3 \ 2 \ 0 \ 2]$$

$$f_3 = 6 \left[\begin{array}{cccccc|c|c} 2 & 2 & 2 & 6 & 0 & 6 & 4 & 6 & 2 & 0 \\ 4 & 1 & 1 & 0 & 2 & 0 & 3 & 5 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 & 5 & 1 & 6 & 2 & 1 \\ 2 & 2 & 2 & 4 & 1 & 1 & 5 & 2 & 2 & 0 \end{array} \right]$$

$$f_3 = 6 \begin{bmatrix} 48 \\ 30 \\ 49 \\ 29 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{matrix} 3 \\ 2 \\ 0 \\ 2 \end{matrix}$$

$$i_3 = 6 \left[\begin{array}{cccccc|c|c} 6 & 5 & 5 & 0 & 1 & 5 & 0 & 1 & 2 & 0 \\ 3 & 2 & 2 & 6 & 2 & 1 & 5 & 6 & 2 & 0 \\ 3 & 2 & 1 & 3 & 0 & 2 & 3 & 2 & 2 & 1 \\ 4 & 4 & 1 & 5 & 4 & 1 & 4 & 0 & 2 & 6 \end{array} \right]$$

$$i_3 = 6 \begin{bmatrix} 47 \\ 46 \\ 27 \\ 48 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{matrix} 3 \\ 2 \\ 0 \\ 2 \end{matrix}$$

$$O_3 = 6 \begin{bmatrix} 5 & 4 & 3 & 5 & 1 & 3 & 2 & 4 \\ 4 & 2 & 2 & 1 & 5 & 0 & 1 & 2 \\ 3 & 0 & 4 & 0 & 2 & 1 & 1 & 5 \\ 4 & 1 & 5 & 3 & 5 & 3 & 3 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix}$$

$$O_3 = 6 \begin{bmatrix} 57 \\ 41 \\ 32 \\ 67 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \\ 2 \end{bmatrix}$$

$$\tilde{C}_3 = \tanh \left[\begin{bmatrix} 0 & 5 & 6 & 0 & 5 & 4 & 0 & 3 \\ 0 & 1 & 5 & 6 & 0 & 3 & 3 & 6 \\ 0 & 0 & 5 & 5 & 0 & 5 & 3 & 4 \\ 2 & 0 & 6 & 5 & 5 & 5 & 6 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right]$$

$$\tilde{C}_3 = \tanh \begin{bmatrix} 57 \\ 46 \\ 38 \\ 63 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \\ 2 \end{bmatrix}$$

$$C_3 = f_3 * C_2 + i_3 * \tilde{C}_3$$

$$C_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

$$h_3 = O_3 * C_3$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

At time = 4

$$I_4 = [h_3, n_4]$$

$$I_4 = [3 \ 3 \ 3 \ 3 \ 1 \ 3 \ 0 \ 2]$$

$$f_4 = 6 \left[\begin{array}{cccccc|c|c} 2 & 2 & 2 & 6 & 0 & 6 & 4 & 6 & 3 & 0 \\ 4 & 1 & 1 & 0 & 2 & 0 & 3 & 5 & 3 & 2 \\ 1 & 2 & 1 & 3 & 4 & 5 & 1 & 6 & 3 & 1 \\ 2 & 2 & 2 & 4 & 1 & 1 & 5 & 2 & 3 & 0 \end{array} \right]$$

$$f_4 = 6 \begin{bmatrix} 66 \\ 32 \\ 53 \\ 38 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}$$

$$i_4 = 6 \left[\begin{array}{cccccc|c|c} 6 & 5 & 5 & 0 & 1 & 5 & 0 & 1 & 3 & 0 \\ 3 & 2 & 2 & 6 & 2 & 1 & 5 & 6 & 3 & 0 \\ 3 & 2 & 1 & 3 & 0 & 2 & 3 & 2 & 3 & 1 \\ 4 & 4 & 1 & 5 & 4 & 4 & 0 & 4 & 3 & 6 \end{array} \right]$$

$$i_4 = 6 \begin{bmatrix} 66 \\ 56 \\ 38 \\ 55 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}$$

$$o_4 = 6 \left[\begin{array}{cccccc|c|c} 5 & 4 & 3 & 5 & 1 & 3 & 2 & 4 & 3 & 6 \\ 4 & 2 & 2 & 1 & 5 & 0 & 1 & 2 & 3 & 4 \\ 3 & 0 & 4 & 0 & 2 & 1 & 1 & 5 & 3 & 0 \\ 4 & 1 & 5 & 3 & 5 & 3 & 3 & 6 & 3 & 8 \end{array} \right] = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}$$

$$C_4 = \tanh \left[\begin{bmatrix} 0 & 5 & 6 & 0 & 5 & 4 & 0 & 3 \\ 0 & 1 & 5 & 6 & 0 & 3 & 3 & 6 \\ 0 & 0 & 5 & 5 & 0 & 5 & 3 & 4 \\ 2 & 0 & 6 & 5 & 5 & 5 & 6 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right]$$

$$C_4 = \tanh \begin{bmatrix} 56 \\ 59 \\ 53 \\ 59 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 3 \\ 0 \\ 2 \end{bmatrix}$$

$$C_4 = f_4 + c_3 + i_4 + \tilde{C}_4$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}$$

$$h_4 = O_4 + C_4$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}$$

Making Prediction

$$\hat{y} = \text{softmax}(W \cdot h_4 + B)$$

$$\hat{y} = \text{softmax} \left[\begin{bmatrix} 6 & 2 & 6 & 0 \\ 1 & 3 & 4 & 1 \\ 5 & 5 & 4 & 1 \\ 6 & 1 & 2 & 1 \\ 4 & 5 & 2 & 3 \\ 2 & 2 & 2 & 5 \\ 4 & 4 & 2 & 6 \\ 1 & 3 & 4 & 6 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \\ 3 \\ 0 \\ 3 \\ 0 \end{bmatrix} \right] = \text{softmax} \begin{bmatrix} 59 \\ 38 \\ 60 \\ 40 \\ 59 \\ 44 \\ 67 \\ 56 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 0.00035 \\ 2.54 \times 10^{-3} \\ 9.1 \times 10^{-4} \\ 1.88 \times 10^{-2} \\ 3.35 \times 10^{-4} \\ 1.02 \times 10^{-10} \\ 0.998 \\ 1.66 \times 10^{-5} \end{bmatrix} \quad \text{Vector} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

→ max →

The students opened their **exams**

USING CONTEXT WORD EMBEDDING

the	=	2 1	0 0	2 1	4	→ n_1
students	=	1	1	1	1	→ n_2
opened	=	2	0	2	2	→ n_3
their	=	0	1	2	1	→ n_4

At time = 1

$$h_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad c_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$I_1 = [h_0, n_1]$$

$$I_1 = [0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 4]$$

$$f_1 = \text{sigmoid}(Wf \cdot I_1 + Bf)$$

$$= 6 \begin{bmatrix} 28 \\ 25 \\ 29 \\ 14 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} = 6 \begin{bmatrix} 28 \\ 27 \\ 30 \\ 14 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_1 = \text{sigmoid}(W_i I_1 + B_i)$$

$$= 6 \begin{bmatrix} 5 \\ 31 \\ 11 \\ 4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} = 6 \begin{bmatrix} 5 \\ 31 \\ 12 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c_1 = \tanh(W_c I_1 + B_c)$$

$$= \tanh \begin{bmatrix} 17 \\ 27 \\ 19 \\ 19 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} = \tanh \begin{bmatrix} 17 \\ 29 \\ 19 \\ 21 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_1 = f_1 * C_0 + i_1 * c_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$O_1 = \text{sigmoid}(W_o I_1 + B_o)$$

$$= \text{sigmoid} \begin{bmatrix} 19 \\ 14 \\ 23 \\ 32 \end{bmatrix} * \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 25 \\ 18 \\ 23 \\ 40 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_1 = O_1 * C_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

At time = 2

$$I_2 = [h_1, n_2]$$

$$I_2 = [1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1]$$

$$f_2 = 6(W_f I_2 + B_f)$$

$$= 6 \left(\begin{pmatrix} 28 \\ 16 \\ 23 \\ 19 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \\ 1 \\ 0 \end{pmatrix} \right) = 6 \begin{pmatrix} 28 \\ 18 \\ 24 \\ 19 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_2 = 6(W_i I_2 + B_i)$$

$$= 6 \left(\begin{pmatrix} 28 \\ 27 \\ 16 \\ 23 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \\ 6 \end{pmatrix} \right) = 6 \begin{pmatrix} 28 \\ 27 \\ 17 \\ 29 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$o_2 = 6(W_o I_2 + B_o)$$

$$= 6 \left(\begin{pmatrix} 27 \\ 17 \\ 16 \\ 30 \end{pmatrix} + \begin{pmatrix} 6 \\ 4 \\ 0 \\ 8 \end{pmatrix} \right) = 6 \begin{pmatrix} 33 \\ 21 \\ 16 \\ 38 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{c}_2 = \tanh(W_c I_2 + B_c)$$

$$= \tanh \left(\begin{pmatrix} 23 \\ 24 \\ 22 \\ 31 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \\ 0 \\ 2 \end{pmatrix} \right) = \tanh \begin{pmatrix} 23 \\ 26 \\ 22 \\ 33 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c_2 = f_2 \cdot c_1 + i_2 \cdot \tilde{c}_2$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

$$h_2 = O_2 \cdot C_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

At time = 3

$$I_3 = [h_2, n_3]$$

$$I_3 = \begin{bmatrix} 2 & 2 & 2 & 2 & 2 & 0 & 2 & 2 \end{bmatrix}$$

$$\begin{aligned} f_3 &= G(W_f \cdot I_3 + B_f) \\ &= G\left(\begin{bmatrix} 44 \\ 32 \\ 26 \\ 36 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix}\right) = G\begin{bmatrix} 44 \\ 34 \\ 37 \\ 36 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} i_3 &= G(W_i I_3 + B_i) \\ &= G\left(\begin{bmatrix} 36 \\ 52 \\ 28 \\ 44 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix}\right) = G\begin{bmatrix} 36 \\ 52 \\ 29 \\ 50 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \tilde{c}_3 &= \tanh(W_c I_3 + B_c) \\ &= \tanh\left(\begin{bmatrix} 38 \\ 42 \\ 34 \\ 52 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix}\right) = \tanh\begin{bmatrix} 38 \\ 44 \\ 34 \\ 54 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} C_3 &= f_3 \cdot c_2 + i_3 \cdot \tilde{c}_2 \\ &= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} \end{aligned}$$

$$o_3 = 6 (W_o I_3 + B_o)$$

$$= 6 \left(\begin{pmatrix} 48 \\ 34 \\ 30 \\ 54 \end{pmatrix} + \begin{pmatrix} 6 \\ 4 \\ 0 \\ 8 \end{pmatrix} \right) = 6 \begin{pmatrix} 54 \\ 38 \\ 30 \\ 62 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_3 = o_3 \cdot c_3$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

At time = 4

$$I_4 = [h_3, n_4]$$

$$I_4 = [3 \ 3 \ 3 \ 3 \ 0 \ 1 \ 2 \ 1]$$

$$f_4 = 6 (W_f I_4 + B_f)$$

$$= 6 \left(\begin{pmatrix} 56 \\ 29 \\ 34 \\ 43 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \\ 1 \\ 0 \end{pmatrix} \right) = 6 \begin{pmatrix} 56 \\ 31 \\ 35 \\ 43 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_4 = 6 (W_i I_4 + B_i)$$

$$= 6 \left(\begin{pmatrix} 54 \\ 56 \\ 37 \\ 51 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \\ 6 \end{pmatrix} \right) = 6 \begin{pmatrix} 54 \\ 56 \\ 38 \\ 57 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$a_4 = 6 (W_o I_4 + B_o)$$

$$= 6 \left(\begin{pmatrix} 62 \\ 31 \\ 29 \\ 54 \end{pmatrix} + \begin{pmatrix} 6 \\ 4 \\ 0 \\ 8 \end{pmatrix} \right) = 6 \begin{pmatrix} 68 \\ 35 \\ 29 \\ 62 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{C}_4 = \tanh(W_c I_4 + B_c)$$

$$= \tanh \left(\begin{pmatrix} 40 \\ 51 \\ 45 \\ 58 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \\ 0 \\ 2 \end{pmatrix} \right) = \tanh \begin{pmatrix} 40 \\ 53 \\ 45 \\ 60 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$C_4 = f_4 \cdot C_3 + i_4 \cdot C_4$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}$$

$$h_4 = o_4 \cdot C_4$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}$$

Making Prediction

$$\hat{y} = \text{softmax} \left(\begin{bmatrix} 56 \\ 36 \\ 60 \\ 40 \\ 56 \\ 44 \\ 64 \\ 56 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \\ 3 \\ 0 \\ 3 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 0.0003 \\ 2.7 \times 10^{-13} \\ 0.00089 \\ 1.9 \times 10^{-12} \\ 0.0003 \\ 1.09 \times 10^{-10} \\ 0.99 \\ 1.7 \times 10^{-5} \end{bmatrix}$$

vector

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

→ max →

The students opened their **exams**.

USING AVERAGE WORD EMBEDDINGS

the =	2	1	2	4
students =	0.5	2	1	2.5
opened =	2.5	1	1	2
their =	0.5	2	1	1.5

At time = 1

$$h_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad c_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$I = [h_0, x_1]$$

$$I = [0 \ 0 \ 0 \ 0 \ 2 \ 1 \ 2 \ 4]$$

$$\begin{aligned} f_1 &= 6(W_f I_1 + B_f) \\ &= 6 \left(\begin{bmatrix} 38 \\ 30 \\ 29 \\ 21 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = 6 \begin{bmatrix} 38 \\ 32 \\ 30 \\ 21 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} i_1 &= 6(W_i I_1 + B_i) \\ &= 6 \left(\begin{bmatrix} 11 \\ 39 \\ 16 \\ 17 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = 6 \begin{bmatrix} 11 \\ 39 \\ 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} o_1 &= 6(W_o T_1 + b_o) \\ &= 6 \left(\begin{bmatrix} 25 \\ 20 \\ 27 \\ 43 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 9 \end{bmatrix} \right) = 6 \begin{bmatrix} 31 \\ 24 \\ 27 \\ 52 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

$$\tilde{c}_1 = \tanh(Wc I_1 + Bc)$$

$$= \tanh\left(\begin{bmatrix} 26 \\ 33 \\ 27 \\ 25 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix}\right) = \tanh\left(\begin{bmatrix} 26 \\ 35 \\ 27 \\ 27 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c_1 = f_1 \cdot c_0 + i_1 \cdot c_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_1 = O_1 \cdot c_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

At time = 2

$$I_2 = [h_1, n_2]$$

$$I_2 = [1 \ 1 \ 1 \ 1 \ 0.5 \ 2 \ 1 \ 2.5]$$

$$f_2 = 6(Wf \cdot I_2 + Bf)$$

$$= 6\left(\begin{bmatrix} 42.9 \\ 28.45 \\ 34.95 \\ 28.42 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix}\right) = 6\begin{bmatrix} 42.9 \\ 24.45 \\ 35.95 \\ 22.42 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

~~42.9~~
~~28.45~~
~~34.95~~
~~22.42~~

$$i_2 = \sigma(W_i I_2 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 28.89 \\ 35.89 \\ 20.93 \\ 21.9 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \begin{bmatrix} 28.89 \\ 35.89 \\ 21.93 \\ 27.9 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{c}_2 = \tanh(W_c I_2 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 28.94 \\ 35.89 \\ 32.9 \\ 36.37 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \begin{bmatrix} 28.94 \\ 37.89 \\ 32.9 \\ 38.37 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c_2 = f_2 \cdot c_1 + i_2 \cdot \tilde{c}_2$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

$$o_2 = \sigma(W_o I_2 + B_o)$$

$$= \sigma \left(\begin{bmatrix} 35.37 \\ 17.43 \\ 23.43 \\ 39.38 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = \sigma \begin{bmatrix} 41.37 \\ 21.43 \\ 23.43 \\ 47.38 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_2 = o_2 \cdot c_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}$$

At time = 3

$$I_3 = [h_2, n_3] \quad \therefore$$

$$I_3 = [2 \ 2 \ 2 \ 2 \ 2.5 \ 1 \ 2 \ 2]$$

$$f_3 = \sigma(W_f I_3 + B_f)$$

$$= \sigma \left(\begin{bmatrix} 45.74 \\ 29.9 \\ 41.85 \\ 32.28 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = \sigma \begin{bmatrix} 45.74 \\ 31.9 \\ 42.85 \\ 32.28 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$i_3 = \sigma(W_i I_3 + B_i)$$

$$= \sigma \left(\begin{bmatrix} 41.14 \\ 48.73 \\ 26.83 \\ 42.74 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = \sigma \begin{bmatrix} 41.14 \\ 48.73 \\ 27.83 \\ 48.74 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\tilde{c}_3 = \tanh(W_c I_3 + B_c)$$

$$= \tanh \left(\begin{bmatrix} 44.16 \\ 41.66 \\ 35.7 \\ 53.14 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \begin{bmatrix} 44.16 \\ 43.66 \\ 35.7 \\ 55.14 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$c_3 = f_3 \cdot c_2 + i_3 \cdot \tilde{c}_3$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

$$O_3 = 6 (W_0 I_3 + B_0)$$

$$= 6 \left(\begin{bmatrix} 49.15 \\ 35.32 \\ 30.81 \\ 56.18 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = 6 \begin{bmatrix} 55.15 \\ 39.32 \\ 30.81 \\ 64.18 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$h_3 = O_3 \cdot c_3$$

$$= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix}$$

At time = 4

$$I_4 = [h_3, n_4]$$

$$I_4 = [3 \ 3 \ 3 \ 3 \ 0.5 \ 2 \ 1 \ 1.5]$$

$$f_4 = 6 (W_f I_4 + B_f)$$

$$= 6 \left(\begin{bmatrix} 60.82 \\ 29.44 \\ 42.91 \\ 40.36 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} \right) = 6 \begin{bmatrix} 60.82 \\ 31.44 \\ 43.91 \\ 40.36 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned}
 i_4 &= 6 (W_i I_4 + B_i) \\
 &= 6 \left(\begin{bmatrix} 59.84 \\ 55.81 \\ 36.89 \\ 49.84 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 6 \end{bmatrix} \right) = 6 \begin{bmatrix} 59.84 \\ 55.81 \\ 37.89 \\ 55.84 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \tilde{c}_4 &= \tanh (W_c I_4 + B_c) \\
 &= \tanh \left(\begin{bmatrix} 47.88 \\ 53.78 \\ 48.8 \\ 60.26 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} \right) = \tanh \begin{pmatrix} 47.88 \\ 55.78 \\ 48.8 \\ 62.26 \end{pmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 c_4 &= f_4 \cdot c_3 + i_4 \cdot \tilde{c}_3 \\
 &= \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 o_4 &= 6 (W_o I_4 + B_o) \\
 &= 6 \left(\begin{bmatrix} 65.29 \\ 33.4 \\ 32.29 \\ 59.3 \end{bmatrix} + \begin{bmatrix} 6 \\ 4 \\ 0 \\ 8 \end{bmatrix} \right) = 6 \begin{bmatrix} 71.29 \\ 37.4 \\ 32.39 \\ 67.3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}
 \end{aligned}$$

$$h_y = 04, C_y = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix}$$

$$\hat{y} = \text{softmax}(W \cdot h_y^T + B)$$

$$= \text{softmax} \left(\begin{bmatrix} 55.82 \\ 35.89 \\ 59.85 \\ 39.88 \\ 55.86 \\ 43.84 \\ 63.8 \\ 55.79 \end{bmatrix} + \begin{bmatrix} 3 \\ 2 \\ 0 \\ 0 \\ 3 \\ 0 \\ 3 \\ 2 \end{bmatrix} \right) = \text{softmax} \begin{bmatrix} 58.82 \\ 37.89 \\ 59.85 \\ 39.88 \\ 58.86 \\ 43.84 \\ 66.8 \\ 57.79 \end{bmatrix}$$

$$= \begin{bmatrix} 0.0003 \\ 2.7 \times 10^{-13} \\ 0.0009 \\ 2 \times 10^{-12} \\ 0.0003 \\ 1 \times 10^{-10} \\ 0.99 \\ 1.2 \times 10^{-4} \end{bmatrix} \xrightarrow{\text{max}} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \text{ vector}$$

The students opened their **exams**.

Results

Whichever embedding we may use whether it be the target, context or average embedding all three of them yield the exact same identical result.

The last word predicted in all three cases is 'exams'.